



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Expanding & Factorising Brackets

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Your notes

Expanding & Simplifying Single Brackets

Expanding One Bracket

How do I expand a bracket?

- The expression $3x(x + 2)$ means $3x$ **multiplied by** the bracket $(x + 2)$
 - $3x$ is the term **outside** the bracket
 - this is sometimes called a **factor**
 - and $x + 2$ are the terms **inside** the bracket
- Expanding the brackets means **multiplying** the **outside term** by **each term** on the **inside**
 - This will **remove** (get rid of) the **brackets**
 - $3x(x + 2)$ expands to $3x \times x + 3x \times 2$ which simplifies to $3x^2 + 6x$
- Beware of **minus signs**
 - Remember the rules
 - $- \times - = +$
 - $- \times + = -$
 - It helps to put **brackets** around **negative** terms



Worked Example

(a) Expand $4x(2x - 3)$.

Multiply the $4x$ term outside the brackets by both terms inside the brackets

$$4x \times 2x + 4x \times (-3)$$

Simplify

$$8x^2 - 12x$$

(b) Expand $-7x(4 - 5y)$.

Multiply the $-7x$ outside the brackets by both terms inside the brackets



Your notes

$$(-7x) \times 4 + (-7x) \times (-5y)$$

Simplify and remember that multiplying two negatives gives a positive

$$-28x + 35xy$$

Expand & Simplify

How do I simplify brackets that are added together?

- First **expand** both brackets **separately**
 - $4(x + 7) + 5x(3 - x)$
 - The **first** set of brackets expands to $4 \times x + 4 \times 7$ which **simplifies** to $4x + 28$
 - The **second** set of brackets expands to $5x \times 3 + 5x \times (-x)$ which **simplifies** to $15x - 5x^2$
 - So $4(x + 7) + 5x(3 - x) = 4x + 28 + 15x - 5x^2$
- Then **collect like terms**
 - $4x + 15x = 19x$
 - The other two terms are **not** like terms
 - So $4(x + 7) + 5x(3 - x) = 19x + 28 - 5x^2$



Worked Example

(a) Expand and simplify $2(x + 5) + 3x(x - 8)$.

Expand each set of brackets separately

You can keep negative terms inside brackets

$$2 \times x + 2 \times 5 + 3x \times x + 3x \times (-8)$$

Simplify each term

$$2x + 10 + 3x^2 - 24x$$

Collect like terms (the $2x$ and the $-24x$)



Your notes

$$-22x + 10 + 3x^2$$

(b) Expand and simplify $3x(x + 2) - 7(x - 6)$.

Expand each set of brackets separately

Be careful: the second set of brackets has a -7 in front, not +7

$$3x \times x + 3x \times 2 + (-7) \times x + (-7) \times (-6)$$

Simplify each term

Remember that multiplying two negatives gives a positive

$$3x^2 + 6x - 7x + 42$$

Collect like terms

$$3x^2 - x + 42$$



Your notes

Expanding Double Brackets

Expanding Two Brackets

How do I expand two brackets using FOIL?

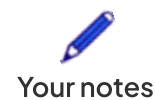
- **Every term** in the **first** bracket must be multiplied by **every term** in the **second** bracket
 - Expanding $(x + 1)(x + 3)$ requires 4 multiplications in total
- A good way to remember all the multiplications is **FOIL**
 - **F = First**: multiply together the **first** terms in each bracket
 - **O = Outside**: multiply the first term in the first bracket by the last term in the last bracket
 - Visually, these are the **outer** terms
 - **I = Inside**: multiply the last term in the first bracket by the first term in the last bracket
 - Visually, these are the **inner** terms
 - **L = Last**: multiply together the **last** terms in each bracket
- It helps to put **negative** terms in **brackets** when multiplying
- **Simplify** the final answer by **collecting like terms** (if there are any)

How do I expand two brackets using a grid?

- You may prefer a more **visual** method using a **grid**
- To expand $(x + 1)(x + 3)$, write out the brackets as **row** and **column headings** of a grid
 - They can be in either direction
 - Remember to write the appropriate **sign** in front of each term

	x	+1
x		
+3		

- For each cell in the grid, **multiply** the term in the **row** heading by the term in the **column** heading



	x	+1
x	x^2	x
+3	$3x$	3

- Add together all the terms **inside** the grid to get the answer

- $x^2 + x + 3x + 3$

- Collect **like** terms

- $x^2 + 4x + 3$

How do I expand when there are multiple variables?

- All the same rules and methods apply as when there is just one variable

- Remember to only simplify **like terms**

- For example: $(3x + 2y)(4x - 6y)$

- Expanding: $12x^2 - 18xy + 8xy - 12y^2$

- The **xy** terms can be combined

- $12x^2 - 10xy - 12y^2$



Worked Example

(a) Expand $(2x - 3)(x + 4)$.

Using FOIL, multiply together the first, outer, inner and last terms

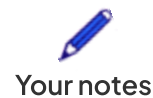
$$\begin{array}{ccccccc}
 & \text{F} & & \text{O} & & \text{I} & & \text{L} \\
 & \text{---} & & \text{---} & & \text{---} & & \text{---} \\
 (2x \times x) & + & (2x \times 4) & + & (-3 \times x) & + & (-3 \times 4)
 \end{array}$$

Simplify each term

$$2x^2 + 8x - 3x - 12$$

Collect like terms (the $8x$ and $-3x$)

$$2x^2 + 5x - 12$$



(b) Expand $(x - 3)(3x - 5)$.

Using FOIL, multiply together the first, outer, inner and last terms

$$\begin{array}{cccc} \text{F} & & \text{O} & & \text{I} & & \text{L} \\ \hline \textcircled{x \times 3x} & + & \textcircled{x \times (-5)} & + & \textcircled{(-3) \times 3x} & + & \textcircled{(-3) \times (-5)} \end{array}$$

Simplify each term

$$3x^2 - 5x - 9x + 15$$

Collect like terms (the $-5x$ and $-9x$)

$$3x^2 - 14x + 15$$



Worked Example

Expand $(3r + 2t)(5t - 8r)$.

Expand using your chosen method, here we will use a grid

	$3r$	$+2t$
$5t$		
$-8r$		

Work out the term in each place in the grid by multiplying

	$3r$	$+2t$
$5t$	$15rt$	$10t^2$
$-8r$	$-24r^2$	$-16rt$

So the expanded expression is

$$10t^2 + 15rt - 16rt - 24r^2$$



Your notes

The ***rt*** terms can be combined

$$10t^2 - rt - 24r^2$$

Expanding Squared Brackets

How do I expand a bracket squared?

- Remember that a **square** number is a number multiplied by itself
- Write $(x + 3)^2$ as $(x + 3)(x + 3)$ and use one of the methods above
 - With FOIL: $(x + 3)(x + 3) = x^2 + 3x + 3x + 9$
 - Then collect like terms: $x^2 + 6x + 9$
- Do **not** make the common **mistake** of saying $(x + 3)^2$ is $x^2 + 3^2$
 - This cannot be true, try substituting in $x = 1$
 - you would get $(1 + 3)^2 = 4^2 = 16$ on the left
 - but $1^2 + 3^2 = 1 + 9 = 10$ on the right



Worked Example

Expand $(2x + 3)^2$.

Remember that the answer is not $(2x)^2 + 3^2$

Rewrite the expression as two separate brackets multiplied together

$$(2x + 3)(2x + 3)$$

Using FOIL, multiply together the first, outer, inner and last terms

F	O	I	L
$(2x \times 2x)$	$+$	$(2x \times 3)$	$+$
$(3 \times 2x)$	$+$	(3×3)	

Simplify each term

$$4x^2 + 6x + 6x + 9$$

Collect like terms (the $6x$ and $6x$)

$$4x^2 + 12x + 9$$



Your notes



Your notes

Expanding Triple Brackets

Expanding Three Brackets

How do I expand three brackets?

- Multiply out any **two of the brackets** using a standard method and **simplify**
- Then multiply the resulting expression by the third (unused) bracket
- This step often looks like $(x + a)(x^2 + bx + c)$
- Every term in the first bracket must be multiplied with every term in the second bracket
- A **grid** can help to keep track of all the terms
 - E.g. $(x + 2)(x^2 + 3x + 1)$

	x^2	$+3x$	$+1$
x	x^3	$3x^2$	x
$+2$	$2x^2$	$6x$	2

- Add all the terms inside the grid together
 - $x^3 + 2x^2 + 3x^2 + 6x + x + 2$
- **Simplify** by collecting any like terms
 - $x^3 + 5x^2 + 7x + 2$



Worked Example

Expand $(2x - 3)(x + 4)(3x - 1)$.

Expand and simplify the first two brackets, for example using the FOIL method



Your notes

$$\begin{aligned}
 & (2x - 3)(x + 4) \\
 &= 2x \times x + 2x \times 4 + (-3) \times x + (-3) \times 4 \\
 &= 2x^2 + 8x - 3x - 12 \\
 &= 2x^2 + 5x - 12
 \end{aligned}$$

Rewrite the original expression with the first two brackets expanded

$$(2x^2 + 5x - 12)(3x - 1)$$

Multiply all of the terms in the first set of brackets by all of the terms in the second set of brackets

A grid can help when there are many terms to multiply together (e.g. write $2x^2 + 5x - 12$ in the vertical column and $3x - 1$ in the horizontal column, then multiply corresponding terms)

	$3x$	-1
$2x^2$	$6x^3$	$-2x^2$
$5x$	$15x^2$	$-5x$
-12	$-36x$	12

Write out the multiplied terms

$$6x^3 - 2x^2 + 15x^2 - 5x - 36x + 12$$

Collect the like terms to simplify

$$6x^3 + 13x^2 - 41x + 12$$



Your notes

Factorising Out Terms

Basic Factorising

What is factorisation?

- A **factorised** expression is one written as the **product (multiplication)** of two, or more, terms (**factors**)
 - $3(x + 2)$ is factorised
 - It is $3 \times (x + 2)$
 - $3x + 6$ is **not** factorised
 - $3xy$ is factorised
 - It is $3 \times x \times y$
 - Numbers can also be factorised
 - $12 = 2 \times 2 \times 3$
- In algebra, factorisation is the **reverse** of **expanding brackets**
 - It's putting it into brackets, rather than removing brackets

How do I factorise two terms?

- To factorise $12x^2 + 18x$
 - Find the highest common factor of the **number** parts
 - 6
 - Find the highest common factor of the **algebra** parts
 - x
 - Multiply both to get the overall **highest common factor**
 - $6x$
 - $12x^2 + 18x$ is the same as $6x \times 2x + 6x \times 3$
 - Using the highest common factor
 - **Take out** the highest common factor
 - Write it **outside** a set of **brackets**



Your notes

- Put the remaining terms, $2x + 3$, inside the brackets
- This gives the answer
 - $6x(2x + 3)$
- To factorise an expression containing multiple variables, e.g. $2a^3b - 4a^2b^2$
 - Use the same approach as above
 - Find the highest common factor of the **number** parts
 - 2
 - Find the highest common factor of the **algebra** parts
 - a and b appear in both terms
 - The highest common factor of a^3 and a^2 is a^2
 - The highest common factor of b and b^2 is b
 - a^2b
 - Multiply both to get the overall **highest common factor**
 - $2a^2b$
 - $2a^3b - 4a^2b^2$ is the same as $2a^2b \times a - 2a^2b \times 2b$
 - Using the highest common factor
 - **Take out** the highest common factor
 - Write it **outside** a set of **brackets**
 - Put the remaining terms, $a - 2b$, inside the brackets
 - This gives the answer
 - $2a^2b(a - 2b)$



Examiner Tips and Tricks

- In the exam, check that your factorisation is correct by **expanding** the brackets!
- **Factorise** mean factorise **fully**.
 - $x(6x + 10)$ is not fully factorised but $2x(3x + 5)$ is.



Your notes

Worked Example

(a) Factorise $5x + 15$

Find the highest common factor of 5 and 15

$$5$$

There is no x in the second term, so no highest common factor in x is needed

Think of each term as $5 \times$ something

$$5 \times x + 5 \times 3$$

Take out the 5 and put $x + 3$ in brackets

$$5(x + 3)$$

$$5(x + 3)$$

(b) Factorise fully $30x^2 - 24x$

Find the highest common factor of 30 and 24

$$6$$

Find the highest common factor of x^2 and x

$$x$$

Find the overall highest common factor by multiplying these together

$$6x$$

Think of each term as $6x \times$ something

$$6x \times 5x - 6x \times 4$$

Take out the $6x$ and put $5x - 4$ in brackets

$$6x(5x - 4)$$

$$6x(5x - 4)$$



Your notes

Factorising by Grouping

Factorising by Grouping

How do I factorise expressions with a common bracket?

- Look at the expression $3x(t + 4) + 2(t + 4)$
 - Both terms have a **common bracket**, $(t + 4)$
 - The whole bracket, $(t + 4)$, can be "taken out" like a common factor:
 - $(t + 4)(3x + 2)$
- This is like factorising $3xy + 2y$ to get $y(3x + 2)$
 - y represents $(t + 4)$ above

How do I factorise by grouping?

- Some questions may require you to **form a common bracket yourself**
 - For example $xy + 3x + 5y + 15$
 - The first two terms have a common factor of x
 - The second two terms have a common factor of 5
 - Factorising fully the **first pair** of terms, and the **last pair** of terms:
 - $x(y + 3) + 5(y + 3)$
 - You can now spot a common bracket of $(y + 3)$
 - $(y + 3)(x + 5)$
- This is called **factorising by grouping**

Does it matter what order I group in?

- You can often **rearrange terms** to factorise in a **different order**
 - Rewriting the same example, $xy + 3x + 5y + 15$, but in a different order:
 - $xy + 5y + 3x + 15$
 - The first pair of terms have a common factor of y
 - The second pair of terms have a common factor of 3



Your notes

- Factorising gives $y(x + 5) + 3(x + 5)$
 - You can now spot a common bracket, this time of $(x + 5)$
- $(x + 5)(y + 3)$
 - This gives the **same result** as found previously
- Some rearrangements **cannot** be factorised as "first pair" then "second pair"
 - For example, rewriting the above example as $xy + 15 + 3x + 5y$



Examiner Tips and Tricks

Once you have factorised something, expand it by hand to check your answer is correct.



Worked Example

Factorise $ab + 3b + 2a + 6$.

Method 1:

Notice that ab and $3b$ have a common factor of b

Notice that $2a$ and 6 have a common factor of 2

Factorise the first two terms, using b as a common factor

$$b(a + 3) + 2a + 6$$

Factorise the second two terms, using 2 as a common factor

$$b(a + 3) + 2(a + 3)$$

$(a + 3)$ is a common bracket

We can now factorise out the bracket $(a + 3)$

$$(a + 3)(b + 2)$$

Method 2:

Notice that ab and $2a$ have a common factor of a

Notice that $3b$ and 6 have a common factor of 3

Rewrite the expression, grouping these terms together

$$ab + 2a + 3b + 6$$

Factorise the first two terms, using a as a common factor

$$a(b + 2) + 3b + 6$$

Factorise the second two terms, using 3 as a common factor

$$a(b + 2) + 3(b + 2)$$

$(b + 2)$ is a common bracket

We can now factorise out the bracket $(b + 2)$

$$(b + 2)(a + 3)$$



Your notes



Your notes

Factorising Simple Quadratics

Factorising Simple Quadratics

What is a quadratic expression?

- A **quadratic expression** is in the form:
 - $ax^2 + bx + c$ (where $a \neq 0$)
- If there are any **higher powers** of x (like x^3 say) then it is **not** a quadratic

How do I factorise quadratics by inspection?

- This is shown most easily through an example: factorising $x^2 - 2x - 8$
- We need a **pair of numbers** that for $x^2 + bx + c$
 - **multiply** to give c
 - which in this case is -8
 - and **add** to give b
 - which in this case is -2
 - $+2$ and -4 satisfy these conditions
 - $2 \times (-4) = -8$ and $2 + (-4) = -2$
 - **Write** these numbers in a **pair of brackets** like this:
 - $(x + 2)(x - 4)$

How do I factorise quadratics by grouping?

- This is shown most easily through an example: factorising $x^2 - 2x - 8$
- We need a **pair of numbers** that for $x^2 + bx + c$
 - **multiply** to give c
 - which in this case is -8
 - and **add** to give b
 - which in this case is -2



Your notes

- +2 and -4 satisfy these conditions
 - $2 \times (-4) = -8$ and $2 + (-4) = -2$
- **Rewrite** the middle term by using +2x and -4x
 - $x^2 + 2x - 4x - 8$
- **Group** and **factorise** the **first two terms**, using x as the common factor
- and **group** and **factorise** the **last two terms**, using -4 as the common factor
 - $x(x + 2) - 4(x + 2)$
- Note that these both now have a **common factor** of (x + 2) so this **whole bracket** can be factorised out
 - $(x + 2)(x - 4)$

How do I factorise quadratics using a grid?

- This is shown most easily through an example: factorising $x^2 - 2x - 8$
- We need a **pair of numbers** that for $x^2 + bx + c$
 - **multiply** to give c
 - which in this case is -8
 - and **add** to give b
 - which in this case is -2
- +2 and -4 satisfy these conditions
 - $2 \times (-4) = -8$ and $2 + (-4) = -2$
- Write the quadratic equation in a **grid** (as if you had used a grid to expand the brackets)
 - splitting the middle term as +2x and -4x
- The grid works by multiplying the row and column headings, to give a product in the boxes in the middle

	x^2	$-4x$
	$+2x$	-8

- Write a heading for the first row, using x as the highest common factor of x^2 and $-4x$

x	x^2	$-4x$
	$+2x$	-8

- You can then use this to find the headings for the columns
 - e.g. "What does x need to be multiplied by to give x^2 ?"
 - and "What does x need to be multiplied by to give $-4x$?"

	x	-4
x	x^2	$-4x$
	$+2x$	-8

- We can then fill in the remaining row heading using the same idea
 - e.g. "What does x need to be multiplied by to give $+2x$?"
 - or "What does -4 need to be multiplied by to give -8 ?"

	x	-4
x	x^2	$-4x$
$+2$	$+2x$	-8

- We can now read off the factors from the column and row headings
 - $(x + 2)(x - 4)$

Which method should I use for factorising simple quadratics?

- The first method, by **inspection**, is by far the **quickest**
 - So this is recommended in an exam for simple quadratics (where $a = 1$)



Your notes

- However the other two methods (grouping, or using a grid) can be used for **harder quadratic equations** where $a \neq 1$

- So you should learn at least one of them too



Examiner Tips and Tricks

As a check, expand your answer and make sure you get the same expression as the one you were trying to factorise.



Worked Example

(a) Factorise $x^2 - 4x - 21$.

We will factorise by inspection

We need two numbers that multiply to give -21, and sum to give -4
+3 and -7 satisfy this

$$3 \times (-7) = -21$$

$$3 + (-7) = -4$$

Write down the brackets

$$(x + 3)(x - 7)$$

(b) Factorise $x^2 - 5x + 6$.

We will factorise by splitting the middle term and grouping

We need two numbers that multiply to 6, and sum to -5
-3 and -2 satisfy this

$$(-3) \times (-2) = 6$$

$$(-3) + (-2) = -5$$

Split the middle term

$$x^2 - 2x - 3x + 6$$



Your notes



Your notes

Factorise x out of the first two terms

$$x(x - 2) - 3x + 6$$

Factorise -3 out of the last two terms

$$x(x - 2) - 3(x - 2)$$

These have a common factor of $(x - 2)$ which can be factored out

$$(x - 2)(x - 3)$$

(c) Factorise $x^2 - 2x - 24$.

We will factorise by using a grid

We need two numbers that multiply to -24 , and sum to -2
 $+4$, and -6 satisfy this

$$4 \times (-6) = -24$$

$$4 + (-6) = -2$$

Use these to split the $-2x$ term and write in a grid

	x^2	$+4x$
	$-6x$	-24

Write a heading using a common factor for the first row

x	x^2	$+4x$
	$-6x$	-24

Work out the headings for the rows

"What does x need to be multiplied by to make x^2 ?"

"What does x need to be multiplied by to make $+4x$?"

	x	$+4$
x	x^2	$+4x$
	$-6x$	-24



Your notes

Repeat for the heading for the remaining row

“What does x need to be multiplied by to make $-6x$?”

(Or “What does $+4$ need to be multiplied by to make -24 ?”)

	x	$+4$
x	x^2	$+4x$
-6	$-6x$	-24

Read off the factors from the column and row headings

$$(x + 4)(x - 6)$$



Your notes

Factorising Harder Quadratics

Factorising Harder Quadratics

How do I factorise a quadratic expression where $a \neq 1$ in $ax^2 + bx + c$?

Method 1: Factorising by grouping

- This is shown most easily through an example: factorising $4x^2 - 25x - 21$
- We need a **pair of numbers** that, for $ax^2 + bx + c$
 - both **multiply** to give ac
 - ac in this case is $4 \times -21 = -84$
 - and both **add** to give b
 - b in this case is -25
- -28 and $+3$ satisfy these conditions
- **Rewrite** the middle term using $-28x$ and $+3x$
 - $4x^2 - 28x + 3x - 21$
- **Group** and fully **factorise** the **first two terms**, using $4x$ as the common factor
- and **group** and fully **factorise** the **last two terms**, using 3 as the common factor
 - $4x(x - 7) + 3(x - 7)$
- These terms now have a **common factor** of $(x - 7)$
 - This **whole bracket** can be factorised out
 - This gives the answer $(x - 7)(4x + 3)$

Method 2: Factorising using a grid

- Use the same example: factorising $4x^2 - 25x - 21$
- We need a **pair of numbers** that for $ax^2 + bx + c$
 - **multiply** to give ac



Your notes

- ac in this case is $4 \times -21 = -84$
- and **add** to give b
 - b in this case is -25
- -28 and $+3$ satisfy these conditions
- Write the quadratic equation in a **grid**
 - (as if you had used a grid to expand the brackets)
 - splitting the middle term up as $-28x$ and $+3x$ (either order)
- The grid works by multiplying the row and column headings, to give a product in the boxes in the middle

	$4x^2$	$-28x$
	$+3x$	-21

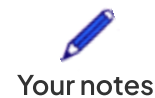
- Write a heading for the first row, using $4x$ as the highest common factor of $4x^2$ and $-28x$

$4x$	$4x^2$	$-28x$
	$+3x$	-21

- You can then use this to find the headings for the columns, e.g. "What does $4x$ need to be multiplied by to give $4x^2$?"

	x	-7
$4x$	$4x^2$	$-28x$
	$+3x$	-21

- We can then fill in the remaining row heading using the same idea, e.g. "What does x need to be multiplied by to give $+3x$?"



	x	-7
$4x$	$4x^2$	$-28x$
$+3$	$+3x$	-21

- We can now read off the brackets from the column and row headings:

- $(x - 7)(4x + 3)$



Worked Example

(a) Factorise $6x^2 - 7x - 3$.

We will factorise by grouping

We need two numbers that:

multiply to $6 \times -3 = -18$

and sum to -7

-9 , and $+2$

Split the middle term up using these values

$$6x^2 + 2x - 9x - 3$$

Factorise $2x$ out of the first two terms

$$2x(3x + 1) - 9x - 3$$

Factorise -3 out of the last two terms

$$2x(3x + 1) - 3(3x + 1)$$

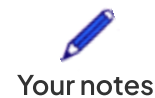
These have a common factor of $(3x + 1)$ which can be factorised out

$$(3x + 1)(2x - 3)$$

(b) Factorise $10x^2 + 9x - 7$.

We will factorise using a grid

We need two numbers that:



multiply to $10 \times -7 = -70$
and sum to $+9$

-5 , and $+14$

Use these values to split the $9x$ term and write in a grid

	$10x^2$	$-5x$
	$+14x$	-7

Write a heading using a common factor of $5x$ from the first row

$5x$	$10x^2$	$-5x$
	$+14x$	-7

Work out the headings for the rows, e.g. "What does $5x$ need to be multiplied by to make $10x^2$?"

	$2x$	-1
$5x$	$10x^2$	$-5x$
	$+14x$	-7

Repeat for the heading for the remaining row, e.g. "What does $2x$ need to be multiplied by to make $+14x$?"

	$2x$	-1
$5x$	$10x^2$	$-5x$
$+7$	$+14x$	-7

Read off the brackets from the column and row headings

$$(2x - 1)(5x + 7)$$



Your notes

Difference of Two Squares

Difference of Two Squares

What is the difference of two squares?

- When a "squared" quantity is subtracted from another "squared" quantity, you get the **difference of two squares**
 - For example:
 - $a^2 - b^2$
 - $9^2 - 5^2$
 - $(x+1)^2 - (x-4)^2$
 - $4m^2 - 25n^2$, which is $(2m)^2 - (5n)^2$

How do I factorise the difference of two squares?

- $a^2 - b^2$ factorises to $(a + b)(a - b)$
 - This can be shown by expanding the brackets
 - $(a + b)(a - b) = a^2 - ab + ba - b^2 = a^2 - b^2$
 - The brackets can swap order
 - $a^2 - b^2 = (a + b)(a - b) = (a - b)(a + b)$
 - (but terms inside a bracket cannot swap order)
- For example, $x^2 - 9 = (x + 3)(x - 3)$
 - This **is the same** as $(x - 3)(x + 3)$
 - But **not the same** as $(3 + x)(3 - x)$
 - which expands to $9 - x^2$

How can the difference of two squares be made harder?

- You may find it used with:
 - numbers**
 - $7^2 - 3^2 = (7+3)(7-3) = (10)(4) = 40$



Your notes

- A combination of **square numbers** and **squared variables**
 - $4m^2 - 9n^2 = (2m)^2 - (3n)^2 = (2m + 3n)(2m - 3n)$
- Any **other powers** which can be written as a **difference of two squares**
 - $a^4 - b^4 = (a^2)^2 - (b^2)^2 = (a^2 + b^2)(a^2 - b^2)$
 - $r^8 - t^6 = (r^4)^2 - (t^3)^2 = (r^4 + t^3)(r^4 - t^3)$
- You may also need to **take out a common factor first**
 - $2x^2 - 18 = 2(x^2 - 9)$ giving $2(x + 3)(x - 3)$
 - The 2 comes out in front

Can I use the difference of two squares to expand?

- Using the difference of two squares to **expand** is **quicker** than expanding double brackets and collecting like terms
- Brackets of the form $(a + b)(a - b)$ **expand to $a^2 - b^2$**
 - For example $(2x + 3)(2x - 3)$ expands to $4x^2 - 9$



Examiner Tips and Tricks

- The difference between two squares is often the trick required to complete a harder algebraic question in the exam
 - Make sure you are able to spot it!



Worked Example

(a) Factorise $9x^2 - 16$.

Recognise that $9x^2$ and 16 are both squared terms

Therefore you can factorise using the difference of two squares

Rewrite as a difference of two squared terms

$$9x^2 - 16 = (3x)^2 - (4)^2$$



Your notes

Use the rule $a^2 - b^2 = (a + b)(a - b)$

$$(3x + 4)(3x - 4)$$

(b) Factorise $4r^2 - t^4$.

Recognise that $4r^2$ and t^4 are both squared terms

Therefore you can factorise using the difference of two squares

Rewrite as a difference of two squared terms

$$4r^2 - t^4 = (2r)^2 - (t^2)^2$$

Use the rule $a^2 - b^2 = (a + b)(a - b)$

$$(2r + t^2)(2r - t^2)$$

(c) Factorise $2y^2 - 50$

This does not appear to be in the form $a^2 - b^2$

There is a common factor of 2, so take this factor out

$$2(y^2 - 25)$$

You can now see $y^2 - 25$ which has the form $y^2 - 5^2$

Use the rule $a^2 - b^2 = (a + b)(a - b)$

$$y^2 - 25 = (y + 5)(y - 5)$$

Now multiply this answer by 2 (leaving the 2 on the outside)

$$2(y + 5)(y - 5)$$



Your notes

Deciding the Factorisation Method

Quadratics Factorising Methods

How do I know if an expression factorises?

- The easiest way to check if $ax^2 + bx + c$ factorises is to check if you can find a pair of integers which:
 - Multiply to give ac
 - Sum to give b
 - If you can find integers to satisfy this, the expression must factorise
- There are some alternate methods to check:
 - Method 1: Use a **calculator** to solve the quadratic expression equal to 0
 - Only some calculators have this functionality
 - If the **solutions are integers or fractions** (without square roots), then the quadratic expression will factorise
 - Method 2: Find the value under the square root in the quadratic formula
 - $b^2 - 4ac$
 - If this number is a **square number**, then the quadratic expression will factorise

Which factorisation method should I use for a quadratic expression?

- Does it have **2 terms only**?
 - Yes, like $x^2 - 7x$
 - Factorise out the highest common factor, x
 - $x(x - 7)$
 - Yes, like $x^2 - 9$
 - Use the "difference of two squares" to factorise
 - $(x + 3)(x - 3)$

Does it have **3 terms**?



Your notes

- Yes, starting with x^2 like $x^2 - 3x - 10$
 - Use "factorising simple quadratics" by finding two numbers that add to -3 and multiply to -10
 - $(x + 2)(x - 5)$
- Yes, starting with ax^2 like $3x^2 + 15x + 18$
 - Check to see if the 3 in front of x^2 is a **common factor** for **all three terms** (which it is in this case), then **factorise it out** of all three terms
 - $3(x^2 + 5x + 6)$
 - The quadratic expression inside the brackets is now $x^2 + \dots$, which factorises more easily
 - $3(x + 2)(x + 3)$
- Yes, starting with ax^2 like $3x^2 - 5x - 2$
 - The 3 in front of x^2 is not a common factor for all three terms
 - Use "factorising harder quadratics", for example factorising by grouping or factorising using a grid
 - $(3x + 1)(x - 2)$

What other expressions should I be able to factorise?

- You may have a **cubed term** like $x^3 - 3x^2 - 10x$
 - Check to see if x is a **common factor** for **all three terms** (which it is in this case), so **factorise it out** of all three terms
 - $x(x^2 - 3x - 10)$
 - The remaining quadratic can then be factorised
 - $x(x + 2)(x - 5)$
- It can also be useful to spot a quadratic in the form $x^2 + 2ax + a^2$
 - This factorises to $(x + a)^2$
 - E.g. $x^2 + 6x + 9 = (x + 3)^2$





Your notes

Examiner Tips and Tricks

- A common **mistake** in the exam is to divide **expressions** by numbers, e.g. $2x^2 + 4x + 2$ becomes $x^2 + 2x + 1$ (which is incorrect)
 - This can only be done with equations
 - e.g. $2x^2 + 4x + 2 = 0$ becomes $x^2 + 2x + 1 = 0$ (dividing "both sides" by 2)



Worked Example

Factorise $-8x^2 + 100x - 48$.

Spot the common factor of -4 and factorise it out

$$-8x^2 + 100x - 48 = -4(2x^2 - 25x + 12)$$

Check to see if the quadratic in the bracket will factorise using $b^2 - 4ac$

$$\begin{aligned} &(-25)^2 - (4 \times 2 \times 12) \\ &= 625 - 96 \\ &= 529 \end{aligned}$$

529 is a square number (23^2) so the expression will factorise

Factorise $2x^2 - 25x + 12$

We require a pair of numbers which multiply to ac , and sum to b

$$a \times c = 2 \times 12 = 24$$

The only numbers which multiply to 24 and sum to -25 are

$$-24 \text{ and } -1$$

Split the $-25x$ term into $-24x - x$

$$2x^2 - 24x - x + 12$$

Group and factorise the first two terms, using $2x$ as the common factor

Group and factorise the last two terms using -1 as the common factor

$$2x(x - 12) - 1(x - 12)$$



Your notes

These factorised terms now have a common term of $(x - 12)$, so this can be factorised out

$$(2x - 1)(x - 12)$$

Recall that -4 was factorised out at the start

$$\begin{aligned} -8x^2 + 100x - 48 &= -4(2x^2 - 25x + 12) = -4(2x - 1)(x - 12) \\ &\quad -4(2x - 1)(x - 12) \end{aligned}$$